

통계적 기계학습

Chapter 1. Orientation

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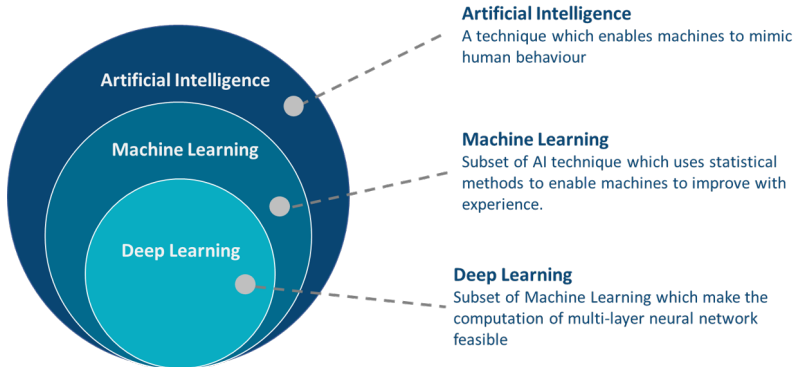
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Statistical learning

We are drowning in information and starving for knowledge.

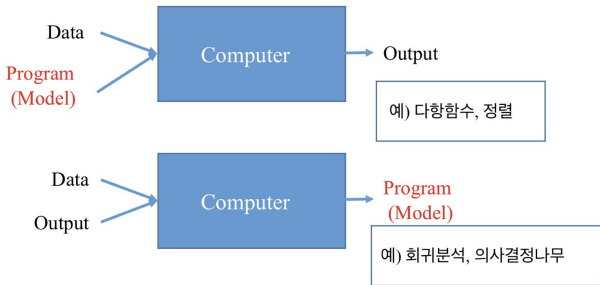
- Rutherford D. Roger

정의 및 개념

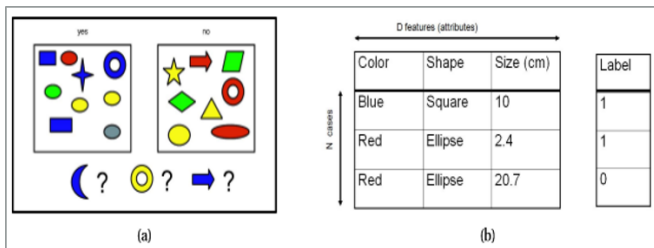


정의 및 개념

- 기계학습(machine learning)이란?
 - 명시적인 프로그램 없이 학습(learning, training) 할 수 있는 방법을 총칭하는 인공지능의 한 분야, 인간이 직접 수많은 규칙을 미리 정해주는 대신 프로그램 자체가 데이터를 통해 스스로 학습하도록 하는 방법
 - Field of study that gives computers the ability to learn without being explicitly programmed. (Arthur Samuel, 1959)



정의 및 개념



(a) : Some labeled training examples of colored shapes, along with 3 unlabeled test cases.

(b) : Representing the training data as an design matrix. The last column is the label.

정의 및 개념

- 학습과제(well-posed learning problem) 중심의 정의
 - A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E . (Tom Mitchell, 1998)
- 학습의 의미?
 - 중요한 패턴과 경향을 추출
 - 데이터가 말하는
- 기계학습은 일종의 함수추정 과정
 - 입력-처리-출력 (함수)
 - 많은, 실제 발생한 (입력-출력) 데이터쌍을 바탕으로 입력을 출력으로 변환하는 함수 추정
 - 종종 출력이 없는 경우

정의 및 개념

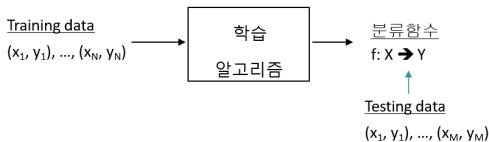
- 28×28 픽셀 크기의 손글씨 이미지 데이터
 - 입력 : N개의 28×28 차원 픽셀값 (0-255)
 - 출력: 0~9까지 숫자 10개



```
00000000010000000000
00000000110000000000
00000000101000000000
00000001000010000000
00000010000010000000
00000100000001000000
00001000000001000000
00001100111111100000
00001111100001000000
00011000000000110000
00010000000000001100
00110000000000000100
00110000000000000100
00100000000000000110
00100000000000000110
01100000000000000010
01000000000000000000
00000000000000000000
```

```
00000000011100000000
00000001100000110000
00000011000000110000
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00000001111000000000
00000011000111000000
00001100000000110000
00001100000000110000
00011000000000011000
00110000000000011000
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- 입력-출력의 함수관계를 학습

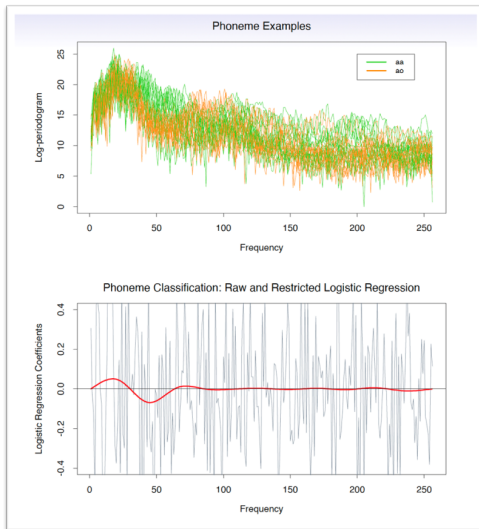


기계학습 사례

- Identify the risk factors for prostate cancer.
- Classify a recorded phoneme based on a log-periodogram.
- Predict whether someone will have a heart attack on the basis of demographic, diet and clinical measurements.
- Customize an email spam detection system.
- Identify the numbers in a handwritten zip code.
- Classify a tissue sample into one of several cancer classes, based on a gene expression profile.
- Establish the relationship between salary and demographic variables in population survey data.
- Classify the pixels in a LANDSAT image, by usage.

From ISL

기계학습 사례



기계학습 사례

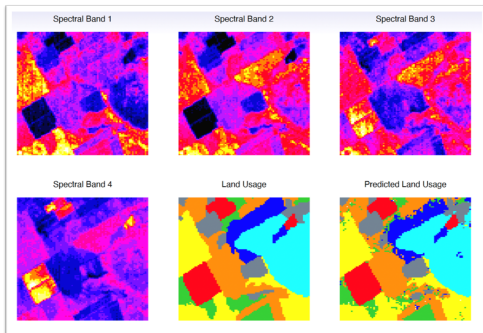
Spam Detection

- data from 4601 emails sent to an individual (named George, at HP labs, before 2000). Each is labeled as *spam* or *email*.
- goal: build a customized spam filter.
- input features: relative frequencies of 57 of the most commonly occurring words and punctuation marks in these email messages.

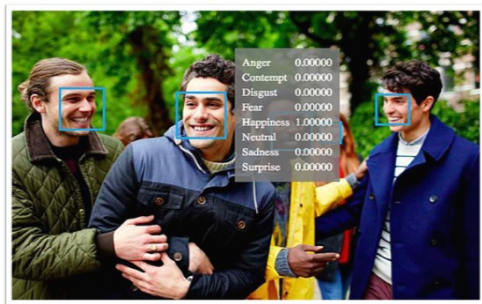
	george	you	hp	free	!	edu	remove
spam	0.00	2.26	0.02	0.52	0.51	0.01	0.28
email	1.27	1.27	0.90	0.07	0.11	0.29	0.01

*Average percentage of words or characters in an email message equal to the indicated word or character. We have chosen the words and characters showing the largest difference between **spam** and **email**.*

기계학습 사례



기계학습 사례



기계학습 사례

- 영상정보로부터 객체 탐지

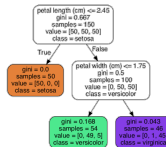
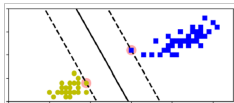
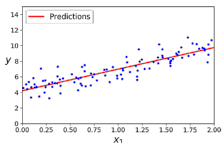


기계학습 위치

- 기계학습 vs 데이터마이닝
 - 유사점 : 자료로부터 중요한 패턴과 경향을 추출
 - 차이점 : 데이터마이닝은 인간의 이해(human comprehension)가 주 목적인 반면, 기계학습은 자료로부터 패턴과 경향을 탐지하여 새로운 자료에 대해 프로그램에 의한 행동을 조정하기 위한 목적
 - * 목적과 수행주체의 차이가 일부 있으나 상당한 중복
- 기계학습 vs 통계적 기계학습(statistical learning)
 - 기계학습은 인공지능의 하위, 통계적 기계학습은 통계학의 하위에서 시작
 - 기계학습은 large scale의 문제에 대해 prediction accuracy에 중점
 - 통계적 기계학습은 uncertainty 고려, model을 통해 interpretability와 precision에 중점
 - * **Statistical learning is a broad term that emphasizes the essential role of statistics within machine learning. But machine learning has the upper hand in marketing.**

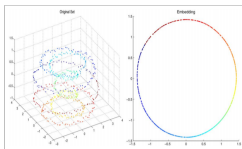
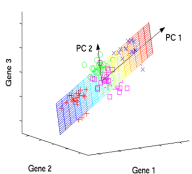
기계학습 구분

- 지도학습(supervised learning)
 - 목표(target, dependent, response) 변수 y 와 설명(explanatory, input, regressors, covariates, features 등) 변수의 벡터 x 가 함께 관찰 (observations, instances 등)
 - 목표변수에 영향을 주는 요인의 식별이나 해석 혹은 목표변수의 예측이 목적
- 지도학습의 구분
 - 목표변수의 형태에 따라 분류(classification)와 회귀(regression)로 구분
 - 분류 : 미리 정의된, 혹은 가능성있는 여러 레이블 중 하나를 예측 혹은 해석
 - 회귀 : 연속성있는 목표변수를 대상



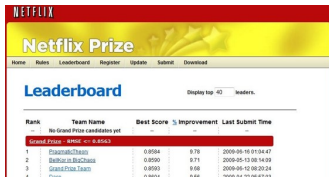
기계학습 구분

- 비지도학습(unsupervised learning)
 - 목표(target, dependent, response) 변수가 관찰되지 않고 설명변수의 벡터만 존재
 - 관측치간 유사한 그룹 혹은 유사한 특징을 갖는 설명변수들을 찾거나, 차원 축소 등
 - 'difficult to know how well you are doing'
 - 'different from supervised learning, but can be useful as a pre-processing step for supervised learning.'



기계학습 구분

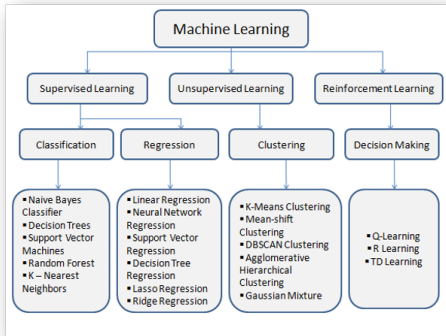
- Netflix prize
 - competition started in October 2006. Training data is ratings for 18,000 movies by 400,000 Netflix customers, each rating between 1 and 5.
 - Training data is very sparse— about 98% missing. objective is to predict the rating for a set of 1 million customer-movie pairs that are missing in the training data.
 - Netix's original algorithm achieved a root MSE of 0.953. The first team to achieve a 10% improvement wins one million dollars.
 - Is this a supervised or unsupervised problem?



The screenshot shows the Netflix Prize Leaderboard interface. At the top, there's a red header with the Netflix logo and a yellow banner for the 'Netflix Prize'. Below the banner is a navigation bar with links: Home, Rules, Leaderboard, Register, Update, Submit, Download. The main section is titled 'Leaderboard' and includes a 'Display top 40 leaders.' link. A table lists the top teams with columns for Rank, Team Name, Best Score, Improvement, and Last Submit Time. The table shows that the 'Grand Prize - RMSE <= 0.8563' is the target, and the top team is 'ChaosTheory' with a score of 0.8584.

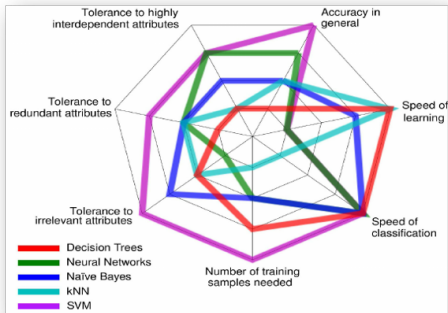
Rank	Team Name	Best Score	Improvement	Last Submit Time
No Grand Prize candidates yet				
Grand Prize - RMSE <= 0.8563				
1	ChaosTheory	0.8584	0.78	2008-06-10 01:04:47
2	Believe in Big Chances	0.8590	0.71	2008-05-13 08:14:39
3	Grand Prize Team	0.8593	0.68	2008-05-12 09:20:24
4	Primo	0.8604	0.68	2008-05-12 08:27:53

기계학습 구분



수많은 기계학습 방법과 알고리즘이 개발

기계학습 구분



언제나 최고의 성능을 보이는
통계적 학습 방법은 존재하지 않음